

# SSC Selection Post (Matriculation Level)

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## High-Yield Quantitative Aptitude + Reasoning Notes

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**Simple rules, fast methods, original examples, and common traps**

**Built from the audited pattern of 40 retained Matriculation-level papers (Phases IX-XII)**

**Important:** These notes teach the question families that appeared repeatedly in the audited papers. Only the official section size is guaranteed. SSC may change topic counts. Reading once is not enough: understand each example, cover its solution, and solve it again without looking.

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### How to use these notes

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For every topic, follow this order:

1. **Recognise** the question family.
2. **Write only the needed formula or diagram.**
3. **Use easy numbers, ratios, or units** instead of long equations whenever possible.
4. **Check the base:** CP or MP? total votes or valid votes? km/h or m/s?
5. **Reject impossible options** before doing lengthy calculation.

### The 40-second rule

If you cannot identify the method within about 40 seconds, mark the question for review and move on. One difficult question should not steal time from five easy ones.

### Symbols used

- CP = Cost Price
- SP = Selling Price
- MP = Marked Price
- P = Principal
- R = Rate percent per year
- T = Time in years
- SI = Simple Interest
- CI = Compound Interest
- D = Distance, S = Speed, T = Time
- LCM = Least Common Multiple
- HCF = Highest Common Factor
- $\pi = 22/7$  or 3.14, as the question requires

# PART A — QUANTITATIVE APTITUDE

## What the paper history says

Across the 40 retained official/reconstructed Phase IX–XII papers, these eight families appeared in **every paper** and formed **876 of 1,000 Maths questions (87.6%)**:

| Priority | Topic                             | Historical items | Observed range per retained paper |
|----------|-----------------------------------|------------------|-----------------------------------|
| 1        | Profit, Loss and Discount         | 197              | 4–6 per paper                     |
| 2        | Percentage                        | 115              | 1–5                               |
| 3        | Simple and Compound Interest      | 111              | 1–3                               |
| 4        | Speed, Distance, Trains and Boats | 107              | 2–4                               |
| 5        | Ratio and Proportion              | 94               | 1–4                               |
| 6        | Average                           | 90               | 1–3                               |
| 7        | Number System and Simplification  | 84               | 1–3                               |
| 8        | Time and Work / Pipes             | 78               | 1–3                               |

**Recent-priority topic:** 3D Mensuration rose to 2–4 questions in every retained Phase XII paper.

**Secondary topics:** 2D Mensuration and Mixture/Alligation.

**Insurance topics:** Partnership, Ages and basic Trigonometry.

**Algebra:** keep algebraic manipulation as a foundation skill; its direct yield fell sharply after Phase IX.

These figures describe the sample; they are not future guarantees.

## 0. Mental toolkit — learn before everything else

### A. Fraction-percentage table

| Fraction      | Percentage        | Fraction       | Percentage       |
|---------------|-------------------|----------------|------------------|
| $\frac{1}{2}$ | 50%               | $\frac{1}{8}$  | 12.5%            |
| $\frac{1}{3}$ | $33\frac{1}{3}\%$ | $\frac{3}{8}$  | 37.5%            |
| $\frac{2}{3}$ | $66\frac{2}{3}\%$ | $\frac{5}{8}$  | 62.5%            |
| $\frac{1}{4}$ | 25%               | $\frac{7}{8}$  | 87.5%            |
| $\frac{3}{4}$ | 75%               | $\frac{1}{10}$ | 10%              |
| $\frac{1}{5}$ | 20%               | $\frac{1}{12}$ | $8\frac{1}{3}\%$ |
| $\frac{2}{5}$ | 40%               | $\frac{1}{16}$ | 6.25%            |
| $\frac{3}{5}$ | 60%               | $\frac{1}{20}$ | 5%               |
| $\frac{4}{5}$ | 80%               | $\frac{1}{25}$ | 4%               |

| Fraction      | Percentage        | Fraction       | Percentage |
|---------------|-------------------|----------------|------------|
| $\frac{1}{6}$ | $16\frac{2}{3}\%$ | $\frac{1}{40}$ | 2.5%       |

## B. Percentage multipliers

- Increase by  $p\%$   $\rightarrow$  multiply by  $(100 + p)/100$ .
- Decrease by  $p\%$   $\rightarrow$  multiply by  $(100 - p)/100$ .
- $x\%$  of  $y = y\%$  of  $x$ .

**Example:**  $16\%$  of  $25 = 25\%$  of  $16 = 4$ .

## C. BODMAS

Always solve in this order:

1. Brackets
2. Orders/powers
3. Division and multiplication, **left to right**
4. Addition and subtraction, **left to right**

## D. Squares worth memorising

$1^2$  to  $20^2$ :

1, 4, 9, 16, 25, 36, 49, 64, 81, 100, 121, 144, 169, 196, 225, 256, 289, 324, 361, 400

Also remember:  $25^2=625$ ,  $30^2=900$ ,  $40^2=1600$ ,  $50^2=2500$ .

## E. The base-100 method

When no absolute amount is given and only a relative percentage or ratio is required, choose a convenient reference base—often 100. It is a working scale, not the actual unknown total.

**20-second recap:** Convert percentages into fractions or multipliers, obey BODMAS, and use base 100 for scale-free relative questions.

# 1. Profit, Loss and Discount

## Why this is first

It was the strongest Maths pattern: **4-6 questions in every one of the 40 retained papers.**

## Recognise it

Look for: bought, cost, sold, gain, loss, marked price, discount, shopkeeper, faulty weight, successive discount.

## Core formulas

$$\text{Profit} = \text{SP} - \text{CP}$$

$$\text{Loss} = \text{CP} - \text{SP}$$

$$\text{Profit\%} = \text{Profit} / \text{CP} \times 100$$

$$\text{Loss\%} = \text{Loss} / \text{CP} \times 100$$

$$\text{SP at } p\% \text{ profit} = \text{CP} \times (100 + p) / 100$$

$$\text{SP at } p\% \text{ loss} = \text{CP} \times (100 - p) / 100$$

$$\text{CP} = \text{SP} \times 100 / (100 + \text{profit\%})$$

$$\text{CP} = \text{SP} \times 100 / (100 - \text{loss\%}) \quad [\text{for loss\% below } 100]$$

$$\text{Discount} = \text{MP} - \text{SP}$$

$$\text{Discount\%} = \text{Discount} / \text{MP} \times 100$$

$$\text{SP after } d\% \text{ discount} = \text{MP} \times (100 - d) / 100$$

## Rule that prevents most mistakes

- Profit/loss percentage is normally calculated on **CP**.
- Discount percentage is calculated on **MP**.

## Type 1 — Direct profit or loss

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**Example:** An item costs ₹800 and is sold at 15% profit. Find SP.

$$\text{SP} = 800 \times 115/100 = ₹920$$

**Reverse example:** An article is sold for ₹920 at 15% profit. Find CP.

$$\text{CP} = 920 \times 100/115 = ₹800$$

## Type 2 — Successive discounts

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Two discounts are applied one after the other, not added directly.

$$\text{Equivalent discount} = d_1 + d_2 - (d_1 d_2 / 100)$$

**Example:** MP = ₹1,500; discounts = 20% and 10%.

Method 1 — multipliers:

$$\text{SP} = 1500 \times 80/100 \times 90/100 = ₹1,080$$

Method 2 — equivalent discount:

$$20 + 10 - (20 \times 10 / 100) = 28\%$$

$$\text{SP} = 1500 \times 72/100 = ₹1,080$$

**Trap:**  $20\% + 10\% \neq 30\%$  because the second discount is applied to the reduced price.

## Type 3 — Markup followed by discount

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Use multipliers.

**Example:** CP = ₹800. A seller marks it 25% above CP and gives 12% discount. Find profit%.

$$\begin{aligned} \text{MP} &= 800 \times 1.25 = 1000 \\ \text{SP} &= 1000 \times 0.88 = 880 \\ \text{Profit} &= 880 - 800 = 80 \\ \text{Profit\%} &= 80/800 \times 100 = 10\% \end{aligned}$$

### One-line method:

$$\text{Net multiplier} = 1.25 \times 0.88 = 1.10 \rightarrow 10\% \text{ profit}$$

## Type 4 — Change in selling price changes profit percent

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If increasing SP by ₹  $x$  changes profit from  $p\%$  to  $q\%$ :

$$\text{CP} = x \times 100 / (q - p)$$

**Example:** An extra ₹90 changes profit from 20% to 35%.

$$\begin{aligned} 15\% \text{ of CP} &= 90 \\ \text{CP} &= 90 \times 100/15 = ₹600 \end{aligned}$$

## Type 5 — Equal SP, one gain and one loss

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If two articles are sold at the **same SP**, one at  $x\%$  gain and the other at  $x\%$  loss:

$$\text{Overall loss\%} = x^2/100$$

**Example:** Same SP; 20% gain on one and 20% loss on the other.

$$\text{Loss\%} = 20^2/100 = 4\%$$

This shortcut works only when both SPs are equal, the gain/loss rates are equal, and  $0 \leq x < 100$ .

## Type 6 — False weight

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If a seller charges the cost of 1,000 g but gives only  $w$  g:

$$\text{Gain\%} = (1000 - w) / w \times 100$$

**Example:** Gives 800 g as 1 kg at the cost-price rate.

$$\text{Gain\%} = 200/800 \times 100 = 25\%$$

## Common traps

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- Calculating profit% on SP instead of CP.
- Adding successive discounts.
- Forgetting that “marked 30% above CP” means  $\text{MP} = 130\% \text{ of CP}$ .
- Using the equal-SP shortcut when the selling prices are not equal.
- Ignoring the weight shortage before applying markup or discount.

**20-second recap:** CP is the base for profit/loss; MP is the base for discount. Convert every change into a multiplier and multiply in order.

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## 2. Percentage

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### Recognise it

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Look for: percent, increase/decrease, income, expenditure, savings, population, votes, marks, pass/fail, consumption, price change.

### Core rules

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$$x\% \text{ of } y = x/100 \times y$$

$$\text{Percentage change} = \text{Change} / \text{Original} \times 100$$

$$\text{New value after } p\% \text{ increase} = \text{Old} \times (100+p)/100$$

$$\text{New value after } p\% \text{ decrease} = \text{Old} \times (100-p)/100$$

### Successive percentage change

For two increases  $a\%$  and  $b\%$ :

$$\text{Net increase}\% = a + b + ab/100$$

For an increase  $a\%$  followed by a decrease  $b\%$ :

$$\text{Net change}\% = a - b - ab/100$$

A negative result means an overall decrease.

### Type 1 — Successive change

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**Example:** A value rises by 20%, then falls by 10%.

$$100 \rightarrow 120 \rightarrow 108$$

$$\text{Net increase} = 8\%$$

Formula:

$$20 - 10 - (20 \times 10 / 100) = 8\%$$

**Trap:** +20% and -20% do not cancel.

$$100 \rightarrow 120 \rightarrow 96 = 4\% \text{ decrease}$$

### Type 2 — “More than” and “less than”

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**Example:** A is 25% more than B.

Assume  $B = 100$ , so  $A = 125$ .

B is less than A by:

$$25/125 \times 100 = 20\%$$

## Fast conversion

If A is  $p\%$  more than B, then B is less than A by:

$$100p / (100+p) \%$$

If A is  $p\%$  less than B, then B is more than A by the following amount, for  $0 \leq p < 100$ :

$$100p / (100-p) \%$$

## Type 3 — Income, expenditure and savings

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Always use:

$$\text{Income} = \text{Expenditure} + \text{Savings}$$

**Example:** Income is ₹20,000. Expenditure is 75% of income. Find savings.

$$\text{Expenditure} = 75\% \text{ of } 20,000 = 15,000$$

$$\text{Savings} = 20,000 - 15,000 = ₹5,000$$

**Example without actual income:** A person spends 80% of income. Income rises 25%, while expenditure rises 10%. Find percentage change in savings.

Assume old income = 100.

$$\text{Old expenditure} = 80; \text{ old saving} = 20$$

$$\text{New income} = 125$$

$$\text{New expenditure} = 80 \times 1.10 = 88$$

$$\text{New saving} = 125 - 88 = 37$$

$$\text{Increase in saving} = 17$$

$$\text{Increase\%} = 17/20 \times 100 = 85\%$$

## Type 4 — Election/votes

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Use the chain:

$$\text{Population or voters} \rightarrow \text{votes cast} \rightarrow \text{valid votes} \rightarrow \text{candidate share}$$

**Example:** 80% of 10,000 voters vote. Of these, 5% are invalid. A gets 60% of valid votes.

$$\text{Votes cast} = 10,000 \times 0.80 = 8,000$$

$$\text{Valid votes} = 8,000 \times 0.95 = 7,600$$

$$\text{A's votes} = 7,600 \times 0.60 = 4,560$$

**Trap:** Candidate percentage is often of valid votes, not total voters.

## Type 5 — Population or repeated growth

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**Example:** Population = 20,000; grows 10% each year for 2 years.

$$\text{New population} = 20,000 \times 1.10^2 = 24,200$$

## Type 6 — Price and consumption

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If price rises and expenditure must stay unchanged:

$$\begin{aligned} & \text{Required reduction in consumption\%} \\ &= \text{Increase\%} / (100 + \text{Increase\%}) \times 100 \end{aligned}$$

**Example:** Price rises 25%.

$$\text{Reduction} = 25/125 \times 100 = 20\%$$

If price falls by  $p\%$ , expenditure stays unchanged, and  $0 \leq p < 100$ , the possible increase in consumption is:

$$p / (100 - p) \times 100\%$$

## Common traps

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- Dividing by the new value instead of the original value.
- Treating successive changes as simple addition.
- Confusing total votes, votes cast and valid votes.
- Forgetting that savings are the difference between income and expenditure.

**20-second recap:** Choose the correct base, convert changes to multipliers, and build percentage chains one step at a time.

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## 3. Simple and Compound Interest

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### Recognise it

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Look for: principal, rate, annual, amount, interest, compounded, half-yearly, instalment, loan, debt.

### Simple Interest

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$$\begin{aligned} \text{SI} &= P \times R \times T / 100 \\ \text{Amount} &= P + \text{SI} \end{aligned}$$

**Example:** ₹2,000 at 6% simple interest for 3 years.

$$\begin{aligned} \text{SI} &= 2000 \times 6 \times 3 / 100 = ₹360 \\ \text{Amount} &= ₹2,360 \end{aligned}$$

### Reverse SI questions

From  $\text{SI} = \text{PRT}/100$  :

$$\begin{aligned} P &= 100 \times \text{SI} / (R \times T) \\ R &= 100 \times \text{SI} / (P \times T) \\ T &= 100 \times \text{SI} / (P \times R) \end{aligned}$$

## Compound Interest

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For compound interest, use a rate and period count that refer to the **same compounding period**:

$$\text{Amount} = P(1 + i)^n$$

$$\text{CI} = \text{Amount} - P$$

Here  $i$  is the decimal rate per compounding period and  $n$  is the number of those periods. For annual compounding,  $i = R/100$  and  $n$  is the number of years.

**Example:** ₹5,000 at 10% compound interest for 2 years.

$$\text{Amount} = 5000 \times 1.10^2 = ₹6,050$$

$$\text{CI} = 6,050 - 5,000 = ₹1,050$$

### Different rates in different years

Multiply each year's factor.

**Example:** ₹10,000 grows by 10% in year 1 and 20% in year 2.

$$\text{Amount} = 10,000 \times 1.10 \times 1.20 = ₹13,200$$

### Half-yearly compounding

$$\text{New rate per period} = R/2$$

$$\text{Number of periods} = 2T$$

**Example:** 12% per year, compounded half-yearly for 1 year:

$$\text{Rate} = 6\% \text{ per half-year; periods} = 2$$

$$\text{Amount} = P \times 1.06^2$$

## Difference between CI and SI

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For exactly 2 years with one unchanged annual rate and annual compounding:

$$\text{CI} - \text{SI} = P(R/100)^2$$

**Example:**  $P = ₹10,000$ ,  $R = 10\%$ ,  $T = 2$  years.

$$\text{Difference} = 10,000 \times (0.10)^2 = ₹100$$

For exactly 3 years with one unchanged annual rate and annual compounding:

$$\text{CI} - \text{SI} = P[3(R/100)^2 + (R/100)^3]$$

## Equal instalments — use a timeline

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Do not guess an instalment formula. Move every payment to the same date.

## Compound-interest example

A loan is cleared by two equal year-end instalments of ₹605 at 10% compound interest. Find the loan.

Bring both payments back to today:

$$\begin{aligned}\text{Loan} &= 605/1.10 + 605/1.10^2 \\ &= 550 + 500 \\ &= ₹1,050\end{aligned}$$

## Simple-interest example

A principal of ₹1,000 is borrowed today at 10% simple interest per annum and is cleared by equal payments at the ends of years 1 and 2. Compare all values at the end of year 2.

At the end of year 2:

$$\begin{aligned}\text{Loan value} &= 1000(1 + 10 \times 2/100) = 1200 \\ \text{First payment } x &\text{ grows for 1 year} \rightarrow 1.10x \\ \text{Second payment } x &\text{ is paid then} \rightarrow x \\ 2.10x &= 1200 \\ x &= ₹571.43 \text{ approximately}\end{aligned}$$

## Common traps

- Using CI formula for SI or vice versa.
- Forgetting to subtract principal when the question asks only for CI.
- For half-yearly compounding, changing the rate but not the number of periods.
- Comparing instalments at different dates.
- Treating “amount” as “interest.”

**20-second recap:** SI grows by the same amount each year; CI grows on the new amount. For instalments, put all values on one date.

# 4. Speed, Distance, Trains and Boats

## Core triangle

$$\begin{aligned}\text{Distance} &= \text{Speed} \times \text{Time} \\ \text{Speed} &= \text{Distance} / \text{Time} \\ \text{Time} &= \text{Distance} / \text{Speed}\end{aligned}$$

## Unit conversion

$$\begin{aligned}\text{km/h to m/s} &\rightarrow \text{multiply by } 5/18 \\ \text{m/s to km/h} &\rightarrow \text{multiply by } 18/5\end{aligned}$$

**Example:**  $72 \text{ km/h} = 72 \times 5/18 = 20 \text{ m/s}$ .

## Average speed

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### Equal distances

If equal distances are travelled at speeds  $u$  and  $v$  :

$$\text{Average speed} = 2uv / (u+v)$$

**Example:** Equal distances at 40 and 60 km/h.

$$\text{Average} = 2 \times 40 \times 60 / 100 = 48 \text{ km/h}$$

### Equal times

$$\text{Average speed} = (u+v)/2$$

**Trap:** Do not use the ordinary average for equal distances.

### General rule

$$\text{Average speed} = \text{Total distance} / \text{Total time}$$

This rule always works. "Total time" means total elapsed time, including any stoppage that the question says to include.

## Relative speed

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- Same direction  $\rightarrow$  relative speed =  $|u-v|$  .
- Opposite directions  $\rightarrow$  relative speed =  $u+v$  .

**Example:** Two runners move in the same direction at 12 m/s and 8 m/s. Relative speed = 4 m/s .

## Trains

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$$\text{Time} = \text{Total length to be crossed} / \text{Relative speed}$$

- Train crossing a pole/person: distance = train length.
- Train crossing a platform: distance = train length + platform length.
- Two trains: distance = sum of train lengths.

**Example:** A 180 m train crosses a 120 m platform at 54 km/h.

$$\text{Speed} = 54 \times 5/18 = 15 \text{ m/s}$$

$$\text{Distance} = 180 + 120 = 300 \text{ m}$$

$$\text{Time} = 300/15 = 20 \text{ seconds}$$

## Boats and streams

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Let boat speed in still water =  $b$  ; stream speed =  $s$  . For upstream travel, the boat must be faster than the stream:  $b > s$  .

$$\text{Downstream speed} = b + s$$

$$\text{Upstream speed} = b - s$$

$$b = (\text{Downstream} + \text{Upstream})/2$$

$$s = (\text{Downstream} - \text{Upstream})/2$$

**Example:** Downstream = 15 km/h; upstream = 9 km/h.

$$\text{Boat speed} = (15+9)/2 = 12 \text{ km/h}$$

$$\text{Stream speed} = (15-9)/2 = 3 \text{ km/h}$$

## Race/chase

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For a faster pursuer following a slower leader on the same route:

$$\text{Catch-up time} = \text{Initial gap} / (\text{Pursuer speed} - \text{Leader speed})$$

**Example:** A is 100 m ahead. B runs 2 m/s faster.

$$\text{Catch-up time} = 100/2 = 50 \text{ seconds}$$

## Common traps

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- Mixing metres with kilometres or seconds with hours.
- Forgetting platform length.
- Adding speeds when objects move in the same direction.
- Using  $(u+v)/2$  for equal-distance journeys.
- Reversing upstream and downstream.

**20-second recap:** Make units equal, decide total distance, decide relative speed, then use  $\text{time} = \text{distance/speed}$ .

# 5. Ratio and Proportion

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## Ratio basics

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$a:b$  means  $a/b$ . Multiplying or dividing both parts by the same non-zero number does not change the ratio.

## Divide a total in a ratio

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If total  $S$  is divided in  $a:b$ :

$$\text{First share} = S \times a/(a+b)$$

$$\text{Second share} = S \times b/(a+b)$$

**Example:** Divide ₹840 in  $3:4$ .

$$\text{First} = 840 \times 3/7 = ₹360$$

$$\text{Second} = 840 \times 4/7 = ₹480$$

## Combine two ratios

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**Example:**  $A:B = 2:3$  and  $B:C = 4:5$ .

Make B equal. LCM of 3 and 4 is 12.

$$A:B = 8:12$$

$$B:C = 12:15$$

$$\text{Therefore } A:B:C = 8:12:15$$

## Proportion

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If:

$$a:b = c:d$$

then:

$$a \times d = b \times c$$

## Mean, third and fourth proportional

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Mean proportional between a and b  $= \sqrt{ab}$

Third proportional to a and b  $= b^2/a$

Fourth proportional to a,b,c  $= bc/a$

**Examples:**

- Mean proportional between 4 and 9  $= \sqrt{36} = 6$ .
- Third proportional to 4 and 6  $= 36/4 = 9$ .
- Fourth proportional to 2, 3, 8  $= 3 \times 8 / 2 = 12$ .

## Direct and inverse proportion

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Use direct proportion only when  $y/x$  stays constant, and inverse proportion only when  $x \times y$  stays constant, with all other relevant factors unchanged.

- In the book example below, cost per book is constant, so cost is directly proportional to quantity.
- In the worker example below, total work and worker efficiency are fixed, so workers and days are inversely proportional.

**Direct example:** 5 books cost ₹300. 8 books cost:

$$300 \times 8 / 5 = ₹480$$

**Inverse example:** 12 workers finish a job in 15 days. 20 equally efficient workers need:

$$12 \times 15 = 20 \times \text{days}$$

$$\text{Days} = 9$$

## Common traps

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- Adding ratio parts to the total instead of using shares.

- Combining ratios without making the common term equal.
- Confusing direct and inverse proportion.
- Using  $ab$  instead of  $\sqrt{ab}$  for mean proportional.

**20-second recap:** Convert ratios into equal units, make common terms equal, and use cross multiplication for proportions.

## 6. Average

### Master formula

Average = Sum / Number of items

Sum = Average × Number of items

### Type 1 — Missing value

**Example:** Average of 5 numbers is 18. Four numbers total 64. Find the fifth.

Total of 5 =  $5 \times 18 = 90$

Fifth =  $90 - 64 = 26$

### Type 2 — Add or remove a member

**Example:** The average age of 8 students is 20 years. A ninth student aged 28 years joins.

Old total =  $8 \times 20 = 160$

New average =  $(160 + 28) / 9 = 188 / 9 = 20 \frac{8}{9}$

### Type 3 — Replacement shortcut

If one value is replaced, preserve the direction of the average change:

New value - Old value =  $n \times (\text{New average} - \text{Old average})$

**Example:** Average of 10 numbers rises by 2 when one number is replaced.

New value - Old value =  $10 \times 2 = 20$

The new number is 20 greater than the old number.

If the average falls, the right-hand side is negative, so the new value is smaller.

### Type 4 — Combined average

Combined average =  $(n_1 a_1 + n_2 a_2) / (n_1 + n_2)$

**Example:** 20 students average 60 marks; 30 students average 70.

Combined average =  $(20 \times 60 + 30 \times 70) / 50$

=  $3300 / 50$

= 66

## Type 5 — Consecutive equally spaced numbers

For an arithmetic sequence, the average is the middle value.

**Example:** Average of 11, 15, 19, 23, 27 is 19.

For an even number of terms, average the two middle terms.

### Common traps

- Adding averages without considering group sizes.
- Forgetting that removal changes both sum and count.
- Using the replacement shortcut without multiplying by the number of items.
- Confusing average speed with the ordinary arithmetic mean.

**20-second recap:** Find the hidden total first:  $\text{sum} = \text{average} \times \text{count}$ . Almost every average problem becomes simple after that.

## 7. Number System and Simplification

### Divisibility rules

| Divisor | Quick test                                                    |
|---------|---------------------------------------------------------------|
| 2       | Last digit is even                                            |
| 3       | Sum of digits is divisible by 3                               |
| 4       | Last two digits are divisible by 4                            |
| 5       | Last digit is 0 or 5                                          |
| 6       | Number is divisible by both 2 and 3                           |
| 8       | Last three digits are divisible by 8                          |
| 9       | Sum of digits is divisible by 9                               |
| 10      | Last digit is 0                                               |
| 11      | Difference of alternating digit sums is 0 or a multiple of 11 |

**Example — divisibility by 11:** Is 2728 divisible by 11?

$$(2+2) - (7+8) = 4-15 = -11$$

Yes, because  $-11$  is a multiple of 11.

### HCF and LCM

For two positive integers:

$$\text{First number} \times \text{Second number} = \text{HCF} \times \text{LCM}$$

**Example:** Product of two numbers = 540; HCF = 6; LCM = 90.

$$\text{HCF} \times \text{LCM} = 6 \times 90 = 540$$

The relation is satisfied.

## Same-remainder questions

The **greatest** divisor that leaves the same unspecified remainder on several numbers is the HCF of their differences.

If the common remainder  $r$  is specified, find:

$$\text{HCF}(\text{number}_1 - r, \text{number}_2 - r, \dots)$$

**Example:** Find the greatest number that divides 65 and 89 leaving the same unspecified remainder.

$$\text{Difference} = 89 - 65 = 24$$

$$\text{Greatest possible divisor} = 24$$

For three or more numbers with an unspecified common remainder, find the HCF of their differences. Other divisors of that HCF may also leave a common remainder, but the HCF gives the greatest one.

## Remainder basics

$$(a+b) \bmod m = [(a \bmod m) + (b \bmod m)] \bmod m$$

$$(a \times b) \bmod m = [(a \bmod m)(b \bmod m)] \bmod m$$

Do not calculate a huge number if only its remainder is needed.

## Unit-digit cycles

| Base ending | Cycle of unit digits | Cycle length |
|-------------|----------------------|--------------|
| 0           | 0                    | 1            |
| 1           | 1                    | 1            |
| 2           | 2, 4, 8, 6           | 4            |
| 3           | 3, 9, 7, 1           | 4            |
| 4           | 4, 6                 | 2            |
| 5           | 5                    | 1            |
| 6           | 6                    | 1            |
| 7           | 7, 9, 3, 1           | 4            |
| 8           | 8, 4, 2, 6           | 4            |
| 9           | 9, 1                 | 2            |

**Example:** Unit digit of  $7^{23}$ .

Cycle length = 4.

$23 \div 4$  leaves remainder 3

Third number in 7-cycle (7,9,3,1) = 3

For a **positive integer exponent**, if division by the cycle length leaves remainder 0, choose the last item in that cycle. For any non-zero base raised to exponent 0, the value—and therefore the unit digit—is 1.

## Simplification

---

- Cancel common factors before multiplication.
- Convert mixed fractions to improper fractions.
- Use BODMAS.
- Keep fractions until the end when decimals become messy.

**Example:**

$$(36 \times 25) / (18 \times 5) = (36/18) \times (25/5) = 2 \times 5 = 10$$

## Common traps

---

- Applying the divisibility-by-4 rule to the whole number instead of its last two digits.
- Forgetting the zero-remainder case in unit-digit cycles.
- Confusing HCF with LCM.
- Solving left to right while ignoring BODMAS.

**20-second recap:** Check divisibility from digits, use differences for same-remainder questions, and use cycles for powers.

# 8. Time and Work / Pipes

---

## Basic idea

---

If A finishes a job in  $x$  days:

$$\text{A's one-day work} = 1/x$$

If A and B work together:

$$\text{Combined rate} = 1/x + 1/y$$

$$\text{Time} = 1 / \text{Combined rate}$$

For two workers only:

$$\text{Together time} = xy/(x+y)$$

## Fast LCM method

---

**Example:** A takes 12 days; B takes 18 days.

Take total work =  $\text{LCM}(12,18) = 36$  units.

$$\text{A's rate} = 36/12 = 3 \text{ units/day}$$

$$\text{B's rate} = 36/18 = 2 \text{ units/day}$$

$$\text{Together} = 5 \text{ units/day}$$

$$\text{Time} = 36/5 = 7.2 \text{ days}$$

## Efficiency and time

---

For the same fixed work completed at constant rates, efficiency is inversely proportional to completion time.

$$\text{Efficiency A} : \text{Efficiency B} = \text{Time B} : \text{Time A}$$

**Example:** A takes 10 days; B takes 15 days.

$$\text{Efficiency A:B} = 15:10 = 3:2$$

## Worker joins or leaves

---

Break the work into periods.

**Example:** A alone takes 10 days. A works for 4 days, then B completes the remaining work in 3 days. Find B's alone time.

$$\text{A does } 4/10 = 2/5$$

$$\text{Remaining} = 3/5$$

$$\text{B does } 3/5 \text{ in 3 days} \rightarrow 1/5 \text{ per day}$$

$$\text{B alone needs 5 days}$$

## Men-days-hours

---

For equally efficient workers operating at constant rates:

$$\text{Workers} \times \text{Days} \times \text{Hours per day} \propto \text{Work}$$

**Example:** 12 workers complete a job in 15 days. How many days for 20 workers?

$$12 \times 15 = 20 \times D$$

$$D = 9 \text{ days}$$

## Pipes and cisterns

---

- Filling pipe = positive work.
- Emptying pipe/leak = negative work.

**Example:** Pipe A fills a tank in 6 h; a leak empties it in 12 h.

$$\text{Net rate} = 1/6 - 1/12 = 1/12$$

$$\text{Tank fills in 12 h}$$

## Alternate-day work

Find work completed in one full cycle, then handle the remainder. Do not count a complete cycle past the finishing point: process the next worker in order and allow a fractional final day if the remaining work is less than that worker's one-day work.

**Example:** A does  $1/6$  per day, B does  $1/12$  per day; they work on alternate days starting with A.

Two-day work =  $1/6 + 1/12 = 1/4$

Three full cycles =  $3/4$  work in 6 days

Day 7: A does  $1/6 \rightarrow$  total  $11/12$

Remaining =  $1/12$

B needs one full day for  $1/12$

Total = 8 days

## Common traps

- Adding completion times instead of work rates.
- Forgetting to subtract a leak's rate.
- Using the LCM as time instead of total work units.
- Ignoring which worker starts an alternate-day schedule.
- Applying inverse proportion when worker efficiencies differ.

**20-second recap:** Convert days into one-day work, add positive rates, subtract negative rates, and use LCM units to avoid fractions.

## 9. 3D Mensuration — recent-priority topic

### Formula table

| Solid            | Volume                 | Curved/Lateral surface | Total surface |
|------------------|------------------------|------------------------|---------------|
| Cube, side $a$   | $a^3$                  | $4a^2$                 | $6a^2$        |
| Cuboid $l, b, h$ | $lbh$                  | $2h(l+b)$              | $2(lb+bh+hl)$ |
| Cylinder         | $\pi r^2 h$            | $2\pi rh$              | $2\pi r(h+r)$ |
| Cone             | $\frac{1}{3}\pi r^2 h$ | $\pi rl$               | $\pi r(l+r)$  |
| Sphere           | $\frac{4}{3}\pi r^3$   | $4\pi r^2$             | $4\pi r^2$    |
| Hemisphere       | $\frac{2}{3}\pi r^3$   | $2\pi r^2$             | $3\pi r^2$    |

For a cone:

Slant height  $l = \sqrt{r^2 + h^2}$

Diagonals:

Cube diagonal =  $a\sqrt{3}$

Cuboid diagonal =  $\sqrt{l^2 + b^2 + h^2}$

## Type 1 — Find a missing dimension

---

**Example:** Cylinder volume =  $1,540 \text{ cm}^3$ , radius = 7 cm, use  $\pi=22/7$ . Find height.

$$1540 = 22/7 \times 7^2 \times h$$

$$1540 = 154h$$

$$h = 10 \text{ cm}$$

## Type 2 — Melting and recasting

---

Material is conserved, so volumes are equal.

$$\text{Old volume} = \text{New volume}$$

**Example:** A cube of side 6 cm is melted into cubes of side 2 cm.

$$\text{Number} = 6^3/2^3 = 216/8 = 27$$

## Type 3 — Water displacement

---

The displaced-water volume equals the volume of the **submerged part** of the object. It equals the whole object's volume only when the object is fully submerged.

For a vertical-sided cuboid tank with no overflow:

$$\text{Submerged volume} = \text{Tank base area} \times \text{Rise in water level}$$

## Type 4 — Change in dimensions

---

If every linear dimension becomes  $k$  times:

$$\text{Area becomes } k^2 \text{ times}$$

$$\text{Volume becomes } k^3 \text{ times}$$

**Example:** Sphere radius doubles.

$$\text{Surface area becomes } 2^2 = 4 \text{ times}$$

$$\text{Volume becomes } 2^3 = 8 \text{ times}$$

## Common traps

---

- Using diameter as radius.
- Confusing curved surface area with total surface area.
- Forgetting the base in total surface area.
- Using area equality in a melting/recasting problem instead of volume equality.
- Forgetting  $1 \text{ m}^3 = 1,000,000 \text{ cm}^3$ .

**20-second recap:** Draw the solid, mark radius/diameter, decide whether the question asks volume, curved area or total area, then substitute units carefully.

---

## 10. Secondary topics

### 10A. 2D Mensuration

| Shape                                                   | Area                                                  | Perimeter/Circumference |
|---------------------------------------------------------|-------------------------------------------------------|-------------------------|
| Rectangle                                               | $lb$                                                  | $2(l+b)$                |
| Square                                                  | $a^2$                                                 | $4a$                    |
| Triangle                                                | $\frac{1}{2} \times \text{base} \times \text{height}$ | Sum of sides            |
| Equilateral triangle                                    | $\sqrt{3}/4 \times a^2$                               | $3a$                    |
| Parallelogram (adjacent sides $a, b$ )                  | $\text{base} \times \text{height}$                    | $2(a+b)$                |
| Trapezium ( $a, b$ parallel; perpendicular height $h$ ) | $\frac{1}{2}(a+b)h$                                   | Sum of sides            |
| Circle                                                  | $\pi r^2$                                             | $2\pi r$                |
| Semicircle                                              | $\frac{1}{2}\pi r^2$                                  | $\pi r + 2r$            |

**Trap:** Perimeter of a semicircle includes the diameter.

#### Path/border questions

- Outer area – inner area = border/path area.
- Draw both dimensions before subtracting.

**Example:** A uniform 2 m wide path runs all around the inside boundary of a  $20 \text{ m} \times 14 \text{ m}$  rectangle.

$$\text{Inner dimensions} = (20-4) \times (14-4) = 16 \times 10$$

$$\text{Path area} = 20 \times 14 - 16 \times 10 = 280 - 160 = 120 \text{ m}^2$$

### 10B. Mixture and Alligation

For two prices or concentrations measured on the same linear basis, and with  $\text{Cheaper} < \text{Mean} < \text{Dearer}$  :

$$\begin{aligned} &\text{Cheaper quantity} : \text{Dearer quantity} \\ &= (\text{Dearer} - \text{Mean}) : (\text{Mean} - \text{Cheaper}) \end{aligned}$$

If the desired mean lies outside the two component values, a positive mixture of only those two components is impossible.

**Example:** Mix rice at ₹30/kg and ₹50/kg to obtain ₹38/kg.

$$\text{Cheaper} : \text{Dearer} = (50-38) : (38-30) = 12:8 = 3:2$$

#### Repeated replacement

If the mixture is homogeneous before every withdrawal, the same  $x$  litres are removed each time from total volume  $V$ , and the vessel is refilled to  $V$  with liquid containing none of the original component being tracked, then after  $n$  repetitions:

$$\text{Original liquid left} = \text{Initial amount} \times (1 - x/V)^n$$

**Example:** From 40 L milk, 10 L is removed and replaced with water twice.

$$\begin{aligned}\text{Milk left} &= 40 \times (1-10/40)^2 \\ &= 40 \times (3/4)^2 \\ &= 22.5 \text{ L}\end{aligned}$$

## Common traps

---

- Reversing the alligation ratio.
- Applying one replacement only when the operation repeats.
- Forgetting that the vessel returns to full volume after each replacement.

**20-second recap:** In alligation, take cross-differences. In replacement, multiply the surviving fraction once for each operation.

---

# Quantitative Aptitude — one-page final revision

---

Percentage:

$$\text{New} = \text{Old} \times (100 \pm p)/100$$

$$\text{Successive increase} = a + b + ab/100$$

$$\text{Increase } a \text{ then decrease } b = a - b - ab/100$$

Profit/Loss:

$$\text{Profit\%} = (\text{SP} - \text{CP})/\text{CP} \times 100$$

$$\text{Loss\%} = (\text{CP} - \text{SP})/\text{CP} \times 100$$

$$\text{SP after discount} = \text{MP}(100 - d)/100$$

$$\text{Two discounts} = d_1 + d_2 - d_1 d_2 / 100$$

Interest:

$$\text{SI} = \text{PRT}/100$$

$$\text{CI amount} = P(1+i)^n, \text{ where } i \text{ is rate per period and } n \text{ is period count}$$

For exactly 2 years at one annual rate, annual compounding:

$$\text{CI} - \text{SI} = P(R/100)^2$$

Speed:

$$D = ST$$

$$\text{km/h} \rightarrow \text{m/s} \times 5/18$$

$$\text{Equal-distance average} = 2uv/(u+v)$$

$$\text{Train time} = \text{total length}/\text{relative speed}$$

$$\text{Downstream} = b + s; \text{ Upstream} = b - s$$

Ratio/Average:

$$\text{Share} = \text{Total} \times \text{part} / \text{sum of ratio}$$

$$\text{Mean proportional} = \sqrt{ab}; \text{ third} = b^2/a; \text{ fourth} = bc/a$$

$$\text{Sum} = \text{Average} \times \text{Count}$$

New value–Old value = Count×(New average–Old average)

Number System:

For two numbers: product=HCF×LCM

Greatest divisor, same unspecified remainder = HCF of differences

Specified remainder  $r$  = HCF of each number– $r$

Use unit-digit cycles for positive integer powers

Work:

One-day work=1/days

Together rate=sum of rates

Leak rate is negative

For equally efficient workers at constant rates:

workers×days×hours per day  $\propto$  work

Mensuration:

Cylinder  $V=\pi r^2 h$

Cone  $V=\frac{1}{3}\pi r^2 h$

Sphere  $V=\frac{4}{3}\pi r^3$

Hemisphere  $V=\frac{2}{3}\pi r^3$

Recasting with no material loss:

old volume = total volume of all new pieces

## Quant attempt order

---

1. Direct percentage, profit/loss, ratio and average.
2. Direct SI, number-system and work questions.
3. Speed/trains and mensuration.
4. Long instalment, multi-stage or unfamiliar questions last.

Do not guess blindly. Check the current official notification: if it states a penalty for wrong answers, a wrong answer costs marks as well as time.

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## PART B — GENERAL INTELLIGENCE / REASONING

---

### What the paper history says

---

Across 40 retained papers, seven broad families appeared in **every paper**; coding-decoding appeared in 39/40.

| Priority | Topic                | Historical items | Observed range per retained paper |
|----------|----------------------|------------------|-----------------------------------|
| 1        | Coding-decoding      | 225              | 0–7                               |
| 2        | Series/sequence      | 178              | <b>3–6 in every paper</b>         |
| 3        | Analogy/relationship | 132              | <b>2–6 in every paper</b>         |

| Priority | Topic                                | Historical items | Observed range per retained paper |
|----------|--------------------------------------|------------------|-----------------------------------|
| 4        | Figural/non-verbal                   | 106              | 2-4 in every paper                |
| 5        | Syllogism/statement logic            | 98               | 2-4 in every paper                |
| 6        | Blood relation                       | 85               | 2-3 in every paper                |
| 7        | Ordering/ranking/arrangement         | 77               | 1-3 in every paper                |
| 8        | Mathematical operations/number logic | 57               | 1-2 in every paper                |
| 9        | Direction/distance                   | 28               | 0-2                               |
| 10       | Classification/odd one out           | 14               | 0-2                               |

The latest retained Phase XII papers were especially regular: 6-7 coding, 5-6 series, 3 analogy, and 2 each from syllogism, figures, blood relation, ordering and operations. Earlier phases differed, so use this as a strong recent pattern—not a guarantee.

## 0. Alphabet toolkit

Many position-based letter questions are easier after converting A-Z to 1-26. First check that the relationship is positional rather than semantic or purely visual.

| A | B | C | D | E | F | G | H | I | J  | K  | L  | M  |
|---|---|---|---|---|---|---|---|---|----|----|----|----|
| 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 | 11 | 12 | 13 |

| N  | O  | P  | Q  | R  | S  | T  | U  | V  | W  | X  | Y  | Z  |
|----|----|----|----|----|----|----|----|----|----|----|----|----|
| 14 | 15 | 16 | 17 | 18 | 19 | 20 | 21 | 22 | 23 | 24 | 25 | 26 |

### Reverse position

Reverse position =  $27 - \text{Normal position}$

Examples:

- Reverse position of C =  $27 - 3 = 24$ .
- Reverse position of T =  $27 - 20 = 7$ .

### Opposite-letter pairs

Opposite letters have positions that add to 27:

A-Z, B-Y, C-X, D-W, E-V, F-U, G-T,  
H-S, I-R, J-Q, K-P, L-O, M-N

## Moving beyond Z

---

Treat the alphabet as a circle:

- $Y + 3 = B$  (  $Y \rightarrow Z \rightarrow A \rightarrow B$  ).
- $B - 4 = X$  (  $B \rightarrow A \rightarrow Z \rightarrow Y \rightarrow X$  ).

## Fast habits

---

- Write positions only for letters that change.
- Test the same rule on every letter before accepting it.
- Look for alternating rules at odd and even positions.

**20-second recap:**  $A=1$ ,  $Z=26$ , opposite positions total 27, and the alphabet wraps around after Z.

# 1. Coding-Decoding

---

## Recognise it

---

Look for: “in a certain code,” “written as,” “coded as,” “what is the code,” or two coded sentences with common words.

## Five-step code detector

---

1. Are the word and code the same length?
2. Test a constant forward/backward shift.
3. Test reversal.
4. Test different rules for odd and even positions.
5. If codes are numbers, test alphabet values, sums, products or positions.

Do not invent a complicated rule until simple rules fail.

## Type 1 — Same shift for every letter

---

**Example:**  $CAT \rightarrow DBU$ . Find the code for  $DOG$ .

$C \rightarrow D (+1)$ ,  $A \rightarrow B (+1)$ ,  $T \rightarrow U (+1)$

$D \rightarrow E$ ,  $O \rightarrow P$ ,  $G \rightarrow H$

Answer: EPH

## Type 2 — Different shift at each position

---

**Example:** A three-letter word is coded by  $+1, +2, +3$ . Code  $BAD$ .

$B+1 = C$

$A+2 = C$

$D+3 = G$

Answer: CCG

Write the shift under each position; do not calculate mentally when signs alternate.

## Type 3 — Reverse, then change

---

**Example rule:** Reverse the word, then move every letter +1. Code `BAT` .

`BAT` → `TAB` → `UBC`

Always perform operations in the stated or discovered order.

## Type 4 — Alphabet-value code

---

**Example:** Code value is the sum of letter positions. Find the value of `BAD` .

`B+A+D` = `2+1+4` = `7`

Possible number rules include:

- Sum of positions
- Difference of first and last
- Product of selected positions
- Number of letters, vowels or consonants
- Reverse alphabet positions

Confirm the rule on every given example.

## Type 5 — Common-code language

---

In this usual puzzle convention, assume each distinct word has one stable, distinct code token and token order is irrelevant, unless the question indicates otherwise. Compare the sets of words and code tokens.

Compare sentences and use common words.

`"red apple sweet"` = `ka mi po`

`"apple is fresh"` = `mi tu na`

The only common word is **apple** and the only common code is **mi**.

`apple` = `mi`

### If two words are common

You may identify only a **set** of two possible words and a set of two possible codes until another sentence separates them. Do not force the individual pairings too early.

## Type 6 — Relabelling

---

If the question says "pen is called book, book is called bag...", answer using the **new name**, not the real name.

**Example:** If water is called milk and milk is called juice, what do we drink to quench thirst?

Real object = water; coded name of water = milk.

**Answer: milk.**

## Common traps

---

- Finding a rule that fits only one letter.
- Forgetting wrap-around after Z.
- Reversing after shifting when the rule requires shifting after reversal.
- Matching a common code without checking that the common word is unique.
- Answering with the real object in a relabelling question.

**20-second recap:** Compare letter positions, test shift/reversal/alternation, and use intersections for sentence codes.

---

## 2. Series and Sequence

---

### Recognise it

---

Look for: next term, missing term, question mark, letter series, number series, or letters to fill several blanks.

### Number-series decision order

---

Check in this order:

1. First differences
2. Second differences
3. Multiplication or division
4. Multiply then add/subtract
5. Squares, cubes, primes or Fibonacci-type sums
6. Odd-position and even-position series separately

### Type 1 — Increasing differences

---

**Example:** 3, 8, 15, 24, 35, ?

Differences: +5, +7, +9, +11

Next difference = +13

$35 + 13 = 48$

### Type 2 — Second difference

---

**Example:** 2, 6, 12, 20, 30, ?

First differences: 4, 6, 8, 10

Next difference: 12

Answer: 42

### Type 3 — Multiplication

---

**Example:** 2, 6, 18, 54, ?

Each term  $\times 3$

Answer = 162

## Type 4 — Multiply and add

---

**Example:** 2, 5, 11, 23, ?

$\times 2 + 1$  each time

$23 \times 2 + 1 = 47$

## Type 5 — Familiar number families

---

Look for:

Squares: 1, 4, 9, 16, 25, 36...

Cubes: 1, 8, 27, 64, 125...

Primes: 2, 3, 5, 7, 11, 13...

Fibonacci: each term = previous two terms added

**Example:** 4, 9, 16, 25, ?  $\rightarrow 2^2, 3^2, 4^2, 5^2$ , so answer  $6^2 = 36$ .

## Type 6 — Two alternating series

---

**Example:** 2, 5, 4, 7, 6, 9, ?

Odd positions: 2, 4, 6, ?  $\rightarrow$  next 8.

Even positions: 5, 7, 9.

**Answer: 8.**

## Letter series

---

Convert letters to positions.

**Example:** A, C, F, J, O, ?

Positions: 1, 3, 6, 10, 15

Differences: +2, +3, +4, +5

Next: +6  $\rightarrow 21 = U$

## Opposite-end series

---

**Example:** AZ, BY, CX, ?

- First letters move forward: A, B, C, **D**.
- Second letters move backward: Z, Y, X, **W**.

**Answer: DW.**

## Gap-filled letter series

---

### Fast method

1. Count the length of a likely repeating block.

2. Divide the line with pencil marks.
3. Complete one block.
4. Check that the same structure works in every block.

**Original example:** a b \_ | a \_ c | \_ b c

Each block should be a b c , so missing letters are c , b , a .

## Common traps

---

- Checking only addition and missing an alternating series.
- Seeing multiplication but ignoring the extra +1 or -1 .
- Converting letters incorrectly after M/N.
- Choosing a rule that fails at an earlier term.
- Filling each blank separately instead of finding the repeated block.

**20-second recap:** Differences first, then ratios, special numbers, and odd/even positions. For letters, convert to 1-26.

---

## 3. Analogy and Relationship

---

### Recognise it

---

Typical form:

A : B :: C : ?

The relationship from A to B must be repeated from C to the answer.

### Type 1 — Meaning relationship

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Common relationships:

- Worker : workplace
- Animal : home
- Tool : function
- Young one : adult
- Part : whole
- Country : capital
- Cause : effect

**Example:** Bird : Nest :: Bee : ?

Nest is the conventional nesting place associated with a bird; a hive is the dwelling/colony structure associated with a bee. **Answer: Hive.**

### Type 2 — Number relationship

---

**Example with two completed pairs:** 8 : 64 , 9 : 81 ; therefore 12 : ?

$$8^2 = 64 \text{ and } 9^2 = 81$$

The simplest common rule is  $n \rightarrow n^2$

$$12^2 = 144$$

**Answer: 144.** In an MCQ, confirm the rule against every completed pair and the options; one pair alone can allow several accidental rules.

Before accepting a rule, test common forms:

- Square/cube
- $\times n \pm k$
- Sum/product of digits
- Reverse digits
- Difference between numbers

## Type 3 — Letter relationship

---

**Example:** CAT : DBU :: DOG : ?

Each letter moves +1, so answer = **EPH**.

## Type 4 — Pair-to-pair relationship

---

When two completed pairs are given before the unknown pair, find the rule common to **both** completed pairs. This removes accidental rules.

## Quick option method

---

Sometimes it is faster to apply the likely rule to the options than to derive a complicated formula from scratch. Eliminate options that fail even one position.

## Common traps

---

- Matching meaning when the question uses letter positions.
- Accepting a numeric rule that works for one pair but not another.
- Reversing the direction: the rule is  $A \rightarrow B$ , not  $B \rightarrow A$ .
- Using number of letters unless the question structure supports it.

**20-second recap:** Say the relationship in one short sentence, then repeat that exact sentence for the second pair.

# 4. Syllogism and Statement-Conclusion

---

## Recognise it

---

The question gives statements such as “All A are B,” followed by conclusions. Treat the statements as true even if they sound strange.

## Draw tiny Venn diagrams

---

This Venn method applies to **categorical** All/No/Some/Only statements. Other statement-conclusion questions require direct analysis of the propositions or evidence.

Use circles; do not rely on real-world knowledge.

All A are B  $\rightarrow A \subseteq B$ ; no A lies outside B

No A is B  $\rightarrow A$  and B do not overlap

Some A are B  $\rightarrow$  place  $\times$  in the overlap

Some A not B  $\rightarrow$  place  $\times$  in A outside B

All A are B does not say that A exists, that A and B differ, or that B has members outside A.

## Four rules to memorise

---

1. **All A are B does not mean all B are A.**
2. **No A is B can be reversed:** no B is A.
3. **Some A are B can be reversed:** some B are A.
4. **Some A are not B cannot be reversed.**

Unless the question explicitly specifies another convention, use **no-existential-import** logic:

- All A are B, No A is B, and Only A are B do not by themselves prove that A or B exists.
- Some A are B provides at least one member in both A and B.
- Some A are not B provides an A-member outside B, but does not prove that any B exists.
- An existential conclusion follows only when a relevant “some” witness is forced through the stated chain into every class named in that conclusion.

## Valid chain patterns

---

### Pattern 1

All A are B

All B are C

Therefore all A are C

### Pattern 2

All A are B

No B is C

Therefore no A is C

### Pattern 3

Some A are B

All B are C

Therefore some A are C

## Pattern 4

Some A are B  
No B is C  
Therefore some A are not C

### Worked example 1

---

Statements:

- All pens are tools.
- All tools are useful.

Conclusion: All pens are useful.

$\text{Pens} \subseteq \text{Tools} \subseteq \text{Useful}$

**Conclusion follows.**

### Worked example 2

---

Statements:

- Some cats are pets.
- No pet is wild.

Conclusion: Some cats are not wild.

The cats marked in the pet circle cannot be wild.

**Conclusion follows.**

### Worked example 3 — the classic trap

---

Statements:

- All roses are flowers.
- Some flowers are red.

Conclusion: Some roses are red.

The red flowers may lie outside the rose circle.

**Conclusion does not follow.**

### Worked example 4

---

Statements:

- No doctor is careless.
- Some teachers are doctors.

Conclusion: Some teachers are not careless.

The teachers who are doctors cannot be careless.

**Conclusion follows.**

## Word “only”

---

For the exact form:

Only A are B = All B are A =  $B \subseteq A$

Do not infer All A are B, and do not infer existence.

**Example:** “Only graduates are eligible” means all eligible people are graduates. It does not mean every graduate is eligible.

Word order matters: A are only B means All A are B. Do not apply this shortcut unchanged to phrases such as “only a few A are B.”

## Conclusion-check method

---

For each conclusion ask:

1. Is it forced in every possible diagram?
2. Did I reverse an “all” statement?
3. Did I create a “some” witness without a relevant “some” premise, or use an unrelated witness?
4. Am I using outside knowledge?

If a conclusion is merely possible but not certain, it does **not** follow in an ordinary definite-conclusion question.

## Common traps

---

- Reversing “all.”
- Treating “some” as “all.”
- Assuming two groups overlap because both are inside a larger group.
- Using facts from real life.
- Confusing “can be” with “must be.”

**20-second recap:** Draw circles. A conclusion follows only if every valid diagram forces it.

## 5. Figural / Non-Verbal Reasoning

---

The archived text layers do not reliably reveal figure subtypes, so these notes teach a general visual method rather than claiming a subtype frequency.

### The SCAN method

---

Check these four things in order:

1. **S — Shape count:** lines, dots, sides, arrows, enclosed regions.
2. **C — Change:** added, removed, shaded or unshaded.
3. **A — Angle/movement:** rotation direction and number of degrees.

4. **N — Next position:** movement around corners, sides or cells.

## Figure series

---

Make a tiny table:

| Feature                       | Figure 1 | Figure 2 | Figure 3 |
|-------------------------------|----------|----------|----------|
| Rotation from previous figure | —        | +90°     | +90°     |
| Dots                          | 1        | 2        | 3        |
| Shading                       | white    | black    | white    |

If Figure 1 starts at 0°, these rotations place Figures 2 and 3 at 90° and 180°; the next transition would add another 90°, giving 270°.

Choose the option that continues **all** changing features, not just one.

## Figure analogy

---

Ask: what exact operation changes the first figure into the second?

Possible operations:

- Rotate 90°, 180° or 270°
- Reflect left-right or top-bottom
- Add/remove one element
- Move an inner symbol clockwise
- Exchange shading
- Combine or subtract shapes

Apply the same operation to the third figure.

## Mirror and water image

---

- Reflection in a **vertical mirror line** swaps left and right while retaining top and bottom.
- Reflection in a **horizontal water line** swaps top and bottom while retaining left and right.
- Reflection is not rotation, and it does not change the count of elements.

## Clock mirror shortcut

For a standard analogue clock reflected across the vertical 12–6 axis, let **t** be the number of minutes after 12:00:

$$\text{Mirror reading} = (720 - t) \bmod 720$$

Display hour 0 as 12. The familiar **11:60 – time** shortcut works only after treating 12 as 0 and normalising the result.

Examples:

$$3:20 \rightarrow 8:40$$

$$3:00 \rightarrow 9:00$$

$$12:20 \rightarrow 11:40$$

11:55 → 12:05

12:00 → 12:00

## Paper folding and holes

---

Undo folds in reverse order. At each crease, reflect only the punch marks carried by the folded flap/layers affected by that fold, and retain only positions on the original sheet.

- An unfold may—but need not—double every visible hole.
- A mark on the crease does not duplicate across that crease.
- Two independent folds can create four positions when all affected layers are punched.

## Embedded/completion figures

---

- Trace one unusual angle or long line first.
- Ignore extra lines in an embedded-figure option.
- In completion, check line direction at every boundary.

## Common traps

---

- Tracking only rotation and ignoring dot count or shading.
- Treating a water image as a mirror image.
- Rotating the page mentally in the wrong direction.
- Rejecting an embedded figure because the option has extra lines.
- Counting decorative intersections as separate objects.

**20-second recap:** Count, compare, track movement, and test every changed feature against the options.

# 6. Blood Relations

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## Draw, do not imagine

---

Use simple symbols:

□ = male

○ = female

— = marriage

| = parent-child

Or write only M, F and arrows if faster.

## Generation levels

---

| Level | Relations            |
|-------|----------------------|
| +2    | Grandparents         |
| +1    | Parents, uncle, aunt |

| Level | Relations                     |
|-------|-------------------------------|
| 0     | Self, sibling, spouse, cousin |
| −1    | Son, daughter, nephew, niece  |
| −2    | Grandchildren                 |

## Paternal and maternal

- Father's side = paternal.
- Mother's side = maternal.

## Simple chain example

- M is the father of N.
- N is the mother of P.

M is P's **maternal grandfather**.

## Coded relation example

Suppose:

$A + B$  means A is the mother of B  
 $A \times B$  means A is the brother of B

Find A's relation to C in  $A + B \times C$ .

A is mother of B.  
 B is brother of C.

If “brother” is defined—or conventionally intended in this puzzle—to mean a full sibling sharing the  
 Without that convention or an extra premise, A's exact relation to C is not determined.

## Pointing questions

Break the sentence from the inside out.

**Example:** A man says, “She is the daughter of the only son of my mother.”

My mother's only son = the speaker  
 Daughter of speaker = his daughter

## “Only” matters

- “Only son” confirms male and excludes other sons; it does not exclude daughters.
- “Only daughter” confirms female and excludes other daughters; it does not exclude sons.
- “Brother's only sister” may be the speaker herself if the speaker is female.

## Unknown gender

---

Do not assume gender from a letter or name. If gender is not stated, the exact relation may be “sibling,” “parent,” or “child” rather than brother/sister, father/mother, son/daughter.

## Coded-chain method

---

1. Rewrite every symbol as a full relation.
2. Mark gender immediately.
3. Place each person by generation.
4. Follow the path between the two people asked.
5. Add maternal/paternal only after identifying the connecting parent.

## Common traps

---

- Starting with the speaker instead of the innermost relation.
- Assuming a spouse is a blood relative.
- Missing gender words.
- Confusing maternal grandfather with paternal grandfather.
- Reading `A is father of B` backwards.

**20-second recap:** Mark gender and generation, draw each link, then trace only the path asked.

---

## 7. Ordering, Ranking and Arrangement

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The ranking formulas below assume one linear order of  $N$  people, unique positions with no ties, and the same population counted from both ends.

### Rank from opposite end

---

For  $N$  people:

$$\text{Rank from other end} = N - \text{given rank} + 1$$

**Example:** Riya is 12th from the left in a row of 30.

$$\text{From right} = 30 - 12 + 1 = 19\text{th}$$

### Total people from both ranks

---

If the same person is  $L$  th from left and  $R$  th from right:

$$\text{Total} = L + R - 1$$

**Example:** 9th from left and 14th from right:

$$\text{Total} = 9 + 14 - 1 = 22$$

## People between two positions

---

If both ranks are measured from the same end:

$$\text{People between} = |\text{Rank}_1 - \text{Rank}_2| - 1$$

**Example:** 7th and 18th from left:

$$18 - 7 - 1 = 10 \text{ people}$$

If ranks come from opposite ends, draw numbered slots unless the order is explicitly clear.

## Dictionary order

---

Compare one letter at a time from the left.

**Example:** Arrange Bat, Batch, Bath, Baton.

- Bat is a complete prefix, so it comes first.
- Compare the next letter after Bat : c, h, o.

Bat, Batch, Bath, Baton

## Dictionary rule

If one word is a complete prefix of another, the shorter word comes first.

## Meaningful order

---

Find the real sequence: beginning → process → result.

**Example:** Seed, fruit, flower, plant.

Seed → Plant → Flower → Fruit

## Simple linear arrangement

---

1. Draw slots.
2. Place absolute clues first: ends, middle, exact positions.
3. Then place relative clues: left/right, next to, between.
4. Recheck every clue.

## Facing direction

- On a north-up diagram, facing north gives left = west and right = east.
- Facing south gives left = east and right = west.

## Common traps

---

- Forgetting  $-1$  because the ranked person is counted from both ends.
- Counting the two named people when asked for people between them.
- Sorting by word length instead of first unequal letter.
- Reversing left/right for a person facing south.

**20-second recap:** Use  $N-r+1$ ,  $L+R-1$ , and difference-1; draw slots when directions mix.

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## 8. Mathematical Operations and Number Logic

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### Symbol substitution

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When symbols have changed meanings:

1. Substitute every operator simultaneously.
2. Preserve operand order and all original parentheses; do not add brackets that change grouping.
3. Evaluate the replacement operators with normal precedence: multiplication/division left to right, then addition/subtraction left to right.

**Example:** If  $+$  means  $\times$  and  $\times$  means  $-$ , evaluate:

$$8 + 3 \times 4$$

Replace signs first:

$$8 \times 3 - 4 = 24 - 4 = 20$$

### Interchanged signs

---

If  $+$  and  $-$  are interchanged, replace every  $+$  with  $-$  and every  $-$  with  $+$  simultaneously. Do not change the new symbols again.

A safe trick is to rewrite them on a new line before calculating.

### Equation correction

---

When asked which signs or numbers should be interchanged:

- Test the easiest-looking option first.
- Rewrite only once per option.
- Use rough size to reject impossible options.
- Apply BODMAS after replacement.

### Missing-number patterns

---

Check simple row/column rules:

- Sum or difference
- Product
- Square or cube
- Sum of squares
- Product  $\pm$  a small number
- Same rule across every row

### Example:

$$2, 3 \rightarrow 13$$

$$4, 5 \rightarrow 41$$

$$6, 7 \rightarrow ?$$

Rule:

$$2^2 + 3^2 = 13$$

$$4^2 + 5^2 = 41$$

$$6^2 + 7^2 = 36 + 49 = 85$$

**Answer: 85.**

## Custom operation

---

If  $a \star b = a^2 + b$ , then:

$$4 \star 3 = 4^2 + 3 = 19$$

Use the definition exactly. A custom symbol does not obey ordinary multiplication rules unless stated. Do not assume ordinary precedence or associativity for repeated custom symbols unless the question defines it.

## Common traps

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- Calculating before replacing all symbols.
- Ignoring BODMAS after replacement.
- Swapping signs twice.
- Finding a pattern that works for only one row.
- Treating a custom symbol like an ordinary mathematical sign.

**20-second recap:** Replace first, calculate second. For number logic, demand one simple rule that works everywhere.

# 9. Direction, Distance and Classification

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These were lower-frequency in the combined archive but are useful insurance topics.

## Coordinate method for direction

---

Use an imaginary graph:

$$\text{North} = +y$$

$$\text{South} = -y$$

$$\text{East} = +x$$

$$\text{West} = -x$$

## Turn table

| Facing | Left turn | Right turn |
|--------|-----------|------------|
| North  | West      | East       |
| East   | North     | South      |
| South  | East      | West       |
| West   | South     | North      |

## Worked example

A person walks 5 m north, turns right and walks 3 m, then turns right and walks 5 m.

Start (0,0)

North 5 → (0,5)

East 3 → (3,5)

South 5 → (3,0)

The person is **3 m east** of the start.

## Shortest distance

If  $x$  and  $y$  are perpendicular net displacements and the question asks for unrestricted straight-line displacement:

Shortest distance =  $\sqrt{(x^2+y^2)}$

If movement is constrained to roads or permitted segments, calculate the shortest **allowed path** instead.

Remember common triples: 3-4-5 , 5-12-13 , 8-15-17 .

## Classification / odd one out

Find the common property shared by three options.

Possible bases:

- Same category or use
- Same number rule
- Same letter shift
- Squares, cubes or primes
- Same figure rotation or symmetry

**Example:** 8, 27, 64, 100

$8=2^3$  ,  $27=3^3$  ,  $64=4^3$

100 is not a cube

**Odd one out: 100.**

## Common traps

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- Reporting total path length instead of shortest displacement.
- Using a weak odd-one-out rule when a clearer common rule exists.
- Applying a turn from the page's orientation instead of the person's current heading. A right quarter-turn is always clockwise, but the resulting compass direction depends on the heading at that moment.

**20-second recap:** Plot movement with coordinates. For classification, state the shared rule of the three matching options.

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## Reasoning — final decision tree

---

When you see a question, identify it before solving:

|                               |                                   |
|-------------------------------|-----------------------------------|
| Coded word/sentence?          | → Coding-decoding                 |
| Terms with ? or blanks?       | → Series                          |
| A:B :: C:?                    | → Analogy                         |
| Categorical All/No/Some/Only? | → Venn syllogism                  |
| Other statement-conclusion?   | → Analyse propositions directly   |
| Only images?                  | → Figural SCAN method             |
| Family words/symbols?         | → Blood relation diagram          |
| Left/right/rank/order?        | → Ranking or arrangement          |
| Changed operators?            | → Rewrite signs, then BODMAS      |
| Walking/turning?              | → Coordinate direction method     |
| Odd option?                   | → Find common rule of other three |

## Reasoning attempt order

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This is a suggested starting order; change it according to your timed accuracy and the difficulty of the actual questions.

1. Direct analogy, ranking and blood relation.
2. Clear series and coding.
3. Syllogism and operations.
4. Figure questions that show an obvious change.
5. Long or unclear pattern questions last.

## Final ten rules

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1. For position-based letter questions, convert letters to positions.
2. Test simple coding rules before complex ones.
3. Split alternating series into odd/even positions.
4. In analogy, repeat the exact relationship.
5. In syllogism, draw circles and accept only forced conclusions.
6. In figures, track count, change, angle and movement.
7. In blood relations, mark gender and generation.

8. In ranking, remember the +1 and -1 corrections.
9. In operator questions, replace all signs before BODMAS.
10. Skip and return if the rule is not visible quickly.

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## Seven-day revision plan

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| Day | Quantitative Aptitude           | Reasoning                       |
|-----|---------------------------------|---------------------------------|
| 1   | Percentage + Profit/Loss basics | Alphabet + Coding               |
| 2   | Discount + Markup               | Number and Letter Series        |
| 3   | SI/CI                           | Analogy + Syllogism             |
| 4   | Ratio + Average                 | Blood Relations + Ranking       |
| 5   | Speed/Trains/Boats              | Operations + Direction          |
| 6   | Time/Work + Pipes               | Figural method + Classification |
| 7   | 3D Mensuration + mixed revision | Mixed revision                  |

Daily method:

1. Read one chapter.
2. Re-solve every worked example without looking.
3. Solve 15–25 questions of that exact family from the retained papers.
4. Record only mistakes and slow methods.
5. Re-solve mistakes after 48 hours.

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## Exam-hall checklist

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Before answering, ask:

- What family is this?
- What is the correct base or unit?
- Is there a direct formula/diagram?
- Can options eliminate work?
- Am I certain enough to risk negative marking?

**The goal is not to solve every question in order. The goal is to collect the easiest accurate marks first.**

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## Evidence and scope

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These notes are original teaching material based on topic frequencies and templates documented in:

- [SSC Matriculation fixed-vs-variable forensic report](#)
- [Detailed Reasoning investigation](#) (*workspace analysis source*)

- [Detailed Maths investigation](#) (*workspace analysis source*)

The audited corpus consists of 40 retained, third-party-hosted historical/reconstructed Matriculation papers from Phases IX–XII. Internal metadata supports level/date consistency but does not authenticate every file as an original SSC release. Historical frequency guides preparation; it cannot guarantee a future topic allocation.